

# JISU JEON

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## RESEARCH INTERESTS

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Low temperature plasma modeling, Plasma driven solution electrochemistry,  
Plasma etching and deposition for microelectronics

## EDUCATION

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| Aug. 2023 – Present   | <b>UNIVERSITY OF MICHIGAN</b><br><i>Ph.D., Nuclear Engineering and Radiological Sciences</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Ann Arbor, USA</b> |
|                       | <ul style="list-style-type: none"><li>• Computational plasma science and engineering group (Principal Investigator: Mark J. Kushner)</li><li>• Formulated plasma liquid chemistries for applications of PFAS remediation and polymerization</li><li>• Relevant courses: Introduction to Low Temperature Plasmas, Plasma Chemistry and Plasma Surface Interactions, Applied Machine Learning (NERS), Machine Learning (EECS, audit)</li></ul>                                                                                                                                                                      |                       |
| Sep. 2021 – May. 2023 | <b>SOGANG UNIVERSITY</b><br><i>M.S., Mechanical Engineering</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Seoul, Korea</b>   |
|                       | <ul style="list-style-type: none"><li>• Won the 4<sup>th</sup> prize among 70+ teams at the 7<sup>th</sup> KSME-SEMES (Korean Society of Mechanical Engineers-Samsung Electronics Mechatronic Solutions) Open Innovation Challenge with a presentation on the development of Monte Carlo collision model considering rotations of diatomic/polyatomic molecules to simulate sputtered film deposition profile</li><li>• Relevant courses: Introduction to Nanotechnology and Engineering, Numerical Heat Transfer, Design of Experiments and Statistical Analyses, Classical Physics, Quantum Mechanics</li></ul> |                       |
| Mar. 2017 – Aug. 2021 | <b>SOGANG UNIVERSITY</b><br><i>B.S., Double major in Mechanical Engineering and Physics</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Seoul, Korea</b>   |
|                       | <ul style="list-style-type: none"><li>• Won the 1<sup>st</sup> prize at Sogang University Student Academic Conference with a presentation on the simulation of magnetic confinement of electrons in the magnetron sputtering reactor</li><li>• Relevant courses: Fluid Mechanics, Continuum Mechanics, Electromagnetism, Heat and Mass Transfer, Finite Element Analysis, Statistical Physics, Modern Optics, Physics Laboratory</li></ul>                                                                                                                                                                        |                       |

## PUBLICATION

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- (In progress) [Jisu Jeon](#), Mark J. Kushner, Aniline Polymerization in Plasma Driven Solution Electrochemistry; expected to be submitted by Feb. 2027
- (In progress) [Jisu Jeon](#), Tiago C. Dias, et al., A Mechanism for Plasma Induced PFAS-in-Water Degradation on Long-Time Scales; expected to be submitted by Dec. 2026
- Tiago C. Dias, [Jisu Jeon](#), et al., Controlling Reactivity at the Plasma-Liquid Interface: Applications to Remediation of PFAS in Water, Plasma Chemistry and Plasma Processing 46, 84 (2026)

## HONORS & AWARDS

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- Second prize, Best PhD Oral Presentation by Organizing Committees' Choice at HAKONE XIX International Symposium on High Pressure Low Temperature Plasma Chemistry, 2026
- Fourth prize, The 7<sup>th</sup> KSME-SEMES Open Innovation Challenge, 2022
- First prize, Sogang University Student Academic Conference, 2020

## SKILLS

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- Python, Matlab, Fortran, C++, Pointwise, COMSOL Multiphysics®